

ROUNSAVILLE TECHNOLOGIES

R-COREX INTELLIGENCE ARCHITECTURE

CODE-TO-CONCEPT MAPPING RECORD

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Version: 1.1.0

Status: Supporting Evidence Record — companion to RT-IP-RCOREX-MASTER-001

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Supersedes: RT-IP-RCOREX-MAPPING-002_v1.0.0 (July 23, 2026), which remains in this archive unmodified as the record of what was true on its own record date.

0. Version History

Version	Date	Change
1.0.0	2026-07-23	Initial mapping against three repositories at RealityStateOS@8e41ae2, presence-resonance@a3f9727, and an empty rounsaville-ai-music-generator.
1.1.0	2026-08-13	Re-run as v1.0.0 Section 8 required. Three findings changed: rounsaville-ai-music-generator is no longer empty; RealityStateOS modules 001 and 004–012 and 999 are no longer unbuilt; and the surveyed scope was too narrow — the account holds ten repositories, not three. Two new IP assets proposed (RT-IP-0016, RT-IP-0017). The presence-resonance finding is re-verified and unchanged.

0.1 Why this is a new version rather than an edit

RT-IP-RCOREX-MAPPING-002_v1.0.0 is dated, checksummed evidence of repository states on July 23, 2026. Every statement in it was accurate on that date. Editing it to reflect later facts would destroy the tamper-evidence this archive exists to provide and would misrepresent what was observed when. v1.0.0 is therefore left byte-identical, its SHA-256 intact, and this document supersedes it — the same pattern RT-IP-RCOREX-MASTER-001 already follows across its v1.0.0 and v1.1.0.

No prior record in this archive was altered in producing this one.

1. Purpose and Method

v1.0.0 Section 8 required that this record be re-run "whenever a named-but-unbuilt module ... is scaffolded, or when rounsaville-ai-music-generator receives its first commit." Both conditions have now occurred. This version discharges that instruction.

Method is unchanged: every repository in scope was cloned and inspected directly — directory structure, README files, source files, test suites, and commit history. Where a concept has no real counterpart, this record says so rather than forcing a mapping. Figures below (test counts, line counts, dependency edges) were measured from the working tree, not quoted from documentation.

One method change: v1.0.0 surveyed three repositories because three were known to it. This version enumerates the account and surveys all ten. See Section 2.2.

2. Repositories Surveyed

2.1 Primary scope (carried forward from v1.0.0)

Repository	Commit Inspected	Visibility	Summary
presencefieldowner-gli tch/RealityStateOS	main-line @ 674d88e	Private	14-module (000–999) verifiable, agent-native operating system. All modules 001–012 and 999 now built and tested — 27 packages install, build, and test in dependency order, 257 tests passing, CI green.
presencefieldowner-gli tch/ronsaville-ai-mus ic-generator	@ afe46fc	Public	No longer empty. Five-layer ecosystem; 003_AUDIO_ENGINE is fully implemented and tested (93 tests). The remaining four layers are declared stubs.
presencefieldowner-gli tch/presence-resonance	main @ a3f9727	Public	Unchanged since v1.0.0. Same commit, same content: all 11 workspace packages still contain only a README and an empty <code>export {}</code> ; . The v1.0.0 finding stands without amendment.

2.2 Repositories outside v1.0.0's scope

v1.0.0's central finding — that R-CoreX is RealityStateOS described abstractly — was reached from a three-repository sample. The account in fact holds ten repositories. Seven were never surveyed:

Repository	Visibility	Status	R-CoreX relevance
ronsaville-ip-protecti on	Public	This archive	Index into RealityStateOS; the only repository that references any other
feathers-of-trinity	Public	Partial	Materially relevant — see Section 6.4

Repository	Visibility	Status	R-CoreX relevance
sonic-presence-flow	Public	Scaffolded	Sensory/AI directory structure overlapping 008_REALITY_TAPE and 005/006
familyville	Public	Scaffolded	Sibling product; overlaps presence-resonance
GlassmorphismAI	Private	Not inspected	Not attached to the compiling session; unverified
RTwebtools	Private	Not inspected	Not attached to the compiling session; unverified
Glassmorphism-ai-0.1.0	Public	No commits	Empty repository

Two repositories could not be read and are recorded as unverified rather than characterized. This record makes no claim about their contents.

3. Central Finding: Unchanged

R-CoreX remains the abstract description of RealityStateOS's own architecture, not a separate codebase. The concept-to-module correspondence in v1.0.0 Section 3 is re-verified and carried forward without amendment. What has changed is that the correspondence is now backed by working code across the whole module map rather than in three modules.

4. Corrected Findings

4.1 rounsaville-ai-music-generator — superseded

v1.0.0 recorded: *"Empty repository. git clone succeeds but returns 'you appear to have cloned an empty repository.' No branches, no files, nothing to map."*

That was accurate on July 23, 2026. It is no longer accurate. The repository now contains a five-layer module ecosystem with one layer fully implemented:

Path	Status	Evidence
003_AUDIO_ENGINE/SynthEngine/	Implemented	Oscillators (sine, square, sawtooth, triangle), scientific-pitch and MIDI conversion in twelve-tone equal temperament, ADSR envelopes, note and sequence rendering. 22 tests.
003_AUDIO_ENGINE/MixMaster/	Implemented	Mixing with clip-avoiding normalization, decibel gain, constant-power panning, stereo interleaving, timeline placement. 24 tests.
003_AUDIO_ENGINE/AudioRenderer /	Implemented	RIFF/WAVE encode and decode, 16-bit PCM and 32-bit IEEE float; decoding walks the chunk list rather than assuming a fixed header offset. 17 tests.
003_AUDIO_ENGINE/PythonRunner/	Implemented	Independent Python implementation of the same synthesis, standard library only. 30 tests.

Path	Status	Evidence
001_FOUNDATION, 002_LLM_GATEWAY, 004_COMPOSITION_AGENT, 005_INTERFACE	Stub	Each package's <code>index.js</code> is <code>module.exports = {}</code> . Declared stubs, not claimed implementations.

Independently verifiable property: SynthEngine (TypeScript/JavaScript) and PythonRunner/synth.py (Python) are separate implementations that produce equivalent audio. Rendering A440 for one second through both with matching parameters yields a maximum per-sample difference of 3.05×10^{-4} — one 16-bit quantization step, the smallest difference representable in the output format.

4.2 RealityStateOS — module completion superseded

v1.0.0 recorded: "Modules 001, 004–012, 999 are named and ordered in the repo's own Build Order but not yet scaffolded."

All are now built with passing tests. Measured dependency edges, read from real relative imports in `.ts` source rather than from the stated Build Order:

Module	Depends on	Tests	Lines
003_FOUNDATION	(nothing)	56	1,527
002_KERNEL	(nothing outside itself)	57	1,014
012_STORAGE	002, 003	10	467
001_BOOTSTRAP	002	15	564
004_VERIFIED_BUS	002, 003	9	391
005_KNOWLEDGE	003, 012	43	1,026
006_REALITY_GRAPH	003, 012	8	284
007_PRESENCE	002, 003, 012	12	477
008_REALITY_TAPE	002, 003, 012	8	368
009_INTELLIGENCE	003, 004	14	536
010_COORDINATION	009	8	313
011_INTERFACE	007	8	459
999_OPERATIONS	001, 002	9	326

003_FOUNDATION/CoreTypes is the most depended-upon component in the portfolio and the location of the Ed25519 and BLAKE3 primitives on which the verification claims throughout the dossier rest.

Scope note carried forward unchanged: 004_VERIFIED_BUS implements SingleNodeConsensus; real multi-node Raft remains documented-as-planned. 011_INTERFACE implements REST and CLI; GraphQL, WebSocket, mobile, desktop, and XR surfaces remain documented-as-planned. 008_REALITY_TAPE excludes hardware sensor intake by design. These limits are stated in the repository itself and are not overstated here.

5. New Material Since v1.0.0

5.1 Verifiable provenance credentials — 005_KNOWLEDGE/src/credentials.ts

The most IP-relevant addition since v1.0.0. 005_KNOWLEDGE previously tracked provenance through a ProvenanceTracker that recorded stage transitions but left them unsigned: any caller could name any actor, and an

entry edited afterward left no trace.

`credentials.ts` converts each stage transition into a signed, chained attestation. A `ProvenanceCredential` binds the claim, the issuer's Ed25519 public key, and the BLAKE3 hash of the preceding credential, then signs a digest over all three. Because each link commits to its predecessor's hash, the chain — not merely each entry — is tamper-evident. Claims are canonicalized (keys sorted, undefined omitted) before hashing, since unstable serialization would render signatures unverifiable.

Verification is opt-in by construction: a verifier with an empty trust set rejects every credential, on the principle that a valid self-consistent signature demonstrates internal integrity but never that the issuer should be believed.

The suite asserts the adversarial cases rather than the happy path: editing a claim breaks the hash; recomputing the hash then fails the signature; re-attributing a credential to another issuer fails; and deleting, reordering, or truncating chain entries breaks the links. `KnowledgeRepository` issues these credentials in the real ingest path.

Documented scope limits: chains are held in memory and are not persisted; issuer private keys have no keystore, rotation, or revocation-list distribution behind them; `revoke()` affects one verifier's local trust set only.

5.2 Continuous integration

RealityStateOS CI now passes end to end for the first time in the run history available at compile time. The blocking defect was structural rather than logical: `003_FOUNDATION/ObjectModel` declared a build script but no test script, so the workflow's `npm install && npm test` loop exited at the seventh of twenty-seven modules under `set -e`, and every module after it went untested on every branch. The module's missing test suite was written and the loop now completes, including the LakeTiticaca Python suite, which had never previously executed in CI.

This matters evidentially: prior to this fix, "tests passing" could be asserted only from local runs. It is now independently reproducible from the repository's own automation.

6. Per-Repository Findings

6.1 RealityStateOS

All v1.0.0 Section 6.1 findings regarding authorship, the Kernel's TypeScript/LakeTiticaca split, the empty-folder redirection for `Scheduler/`, `SecurityKernel/`, `PolicyEngine/`, and the spec-only status of `Memory/`, `ProcessManager/`, `ThreadManager/` are re-verified and carried forward. The Kernel-to-LakeTiticaca bridge remains unbuilt and remains explicitly deferred to `004_VERIFIED_BUS`.

6.2 rounsaville-ai-music-generator — superseded

See Section 4.1. v1.0.0's instruction to "re-run this mapping exercise once code is pushed" is hereby discharged.

6.3 presence-resonance — unchanged

Re-verified at the same commit `a3f9727` with identical content. The v1.0.0 assessment stands in full: it is a sibling product sharing only common authorship, the `packages/resonance` and `packages/ai-interpreter` names remain coincidental echoes of SRI and IHI, both remain empty `export {}`; stubs, and citing this repository as evidence of SRI or IHI reduction-to-practice would still not be supportable.

6.4 feathers-of-trinity — a second LakeTiticaca

This repository was outside v1.0.0's scope and is materially relevant, because it contains a **second implementation of a named R-CoreX asset**.

RT-IP-0015 (LakeTiticaca — Kernel AI Cognition Loop) is registered against `RealityStateOS/002_KERNEL/LakeTiticaca`. A separate `ecosystem/laketiticaca_interpreter.py` exists in

feathers-of-trinity, with no code link between the two. The repository additionally carries its own ECOSYSTEM.md architecture tree using an implemented/partial/scaffolded/vision status vocabulary, and its core/runtime-kernel package is independently implemented and tested.

For registry purposes: an asset implemented twice across two repositories should have both locations recorded, or one should be designated canonical. Leaving the duplication undocumented risks an inconsistent account of where the invention is actually reduced to practice.

6.5 Portfolio-level observation: no cross-repository code coupling

Across all eight readable repositories, no repository imports another. No shared packages, no workspace links, no references by name outside this archive. Every relationship in the portfolio — including every mapping asserted in this record — is documentary rather than executable.

This does not weaken any individual claim; each repository stands on its own code. It does mean the portfolio is a set of independent products under common authorship, not an integrated system, and it should be described that way.

7. IP Registry Additions

Proposed for RT-IP-RCOREX-MASTER-001 Section 13, additional to RT-IP-0014 and RT-IP-0015 from v1.0.0:

asset-index.csv has been maintained ahead of this record: it already carries RT-IP-0016 through RT-IP-0026, one row per RealityStateOS module, marked implemented. Those rows are consistent with Section 4.2's measurements and are left as they stand. The next free identifiers are therefore RT-IP-0027 onward.

ID	Asset	Basis
RT-IP-0027	Verifiable Provenance Credential Chain	005_KNOWLEDGE/src/credentials.ts — Ed25519-signed, BLAKE3-hash-chained provenance attestations over a staged pipeline, with canonical serialization and opt-in issuer trust. Implemented and adversarially tested.
RT-IP-0028	Dual-Implementation Audio Synthesis Engine	rounsaville-ai-music-generator/003_AUDIO_ENGINE — independent JavaScript and Python synthesis implementations, measured equivalent to one 16-bit quantization step.

Both rows are added to asset-index.csv by this revision, bringing it to 28 assets.

RT-IP-0015 should be amended to record the second implementation site identified in Section 6.4. RT-IP-0018 (005_KNOWLEDGE) now additionally carries the credential layer described in Section 5.1.

8. Limitations

This record establishes technical provenance and a documented mapping between conceptual architecture and real source code, as of the repository states inspected on August 13, 2026. It is not a substitute for legal advice, and it does not itself establish patent, trademark, or copyright rights — see Section 27 of RT-IP-RCOREX-MASTER-001 for the governing limitations, which apply equally here.

Two repositories (GlassmorphismAI, RTwebtools) were not readable by the compiling session and are recorded as unverified; any claim over their contents requires a separate survey.

At compile time, work referencing a 013_TELECOMMUNICATIONS module and a 002_KERNEL/Telecommunications submodule was observed in a working environment outside this session. Neither exists on any branch of any remote

reachable at compile time, so neither could be inspected and neither is mapped here. **Unpushed work cannot be evidenced by this archive.** If that module is intended as an IP asset, it must be committed and pushed before a record can cite it.

Repository states change. This record should be re-run when the Kernel-to-LakeTiticaca bridge is built, when any documented-as-planned surface (Raft consensus, GraphQL/WebSocket/XR interfaces) is implemented, when the two unverified private repositories become readable, or when 013_TELECOMMUNICATIONS is pushed.

END OF MAPPING RECORD